Histogram

PDF, CDF, mean, variance, std deviation

Random Variable

$x = 2$

$Y = 3$

$\boxed{2}$

x

tossing a coin $\begin{cases} H \\ T \end{cases}$

$x = H$

$x = T$

Random variable $\rightarrow$ $x = \{H, T\}$

Dice

$x = [1, 2, 3, 4, 5, 6]$

Rand

Random variable

$x = \{1, 2, 3, 4, 5, 6\}$

(i) C.R.V

Discrete Random variable

(ii) D.R.V

Random variable

C.R.V

Visited

classroom

$h \in$ Real

$h = \{165, 155 \ldots\}$

(∞ chance)

Dx
C1
salary
C2
designation
p.r.v
A
B
list → finite
[- - - -]
categorical
C1 ∈ Real
C r.v

Tossing a coin

visited $h = - - - - - - -$

$P(h > 160) = ?$

$S = \{H, T\}$

D.R.V

Prob $\rightarrow$ $P(x = H) = \frac{1}{2}$ | $P(x = T) = \frac{1}{2}$

$\underbrace{\qquad\qquad}_{1}$

Sample space

$x = \{HH, HT, TH, TT\}$

$P(x=0) = \left(\frac{1}{4}\right)$

x	0	1	2
$P(x=x)$	$1/4$	$2/4$	$1/4$
	$1/4$	$3/4$	1

$\sum_{i=1}^{n} P(x=x) = 1$

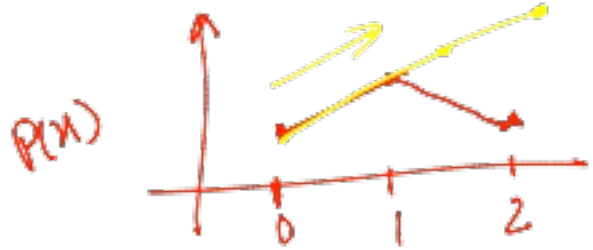
$\frac{4}{4} \Rightarrow 1$

D.R.V

P.m.f

$\boxed{1}$

Prob distribution table



Population & sample

1B → earth → task

Avg hi ⇒ [- , - , - , -]

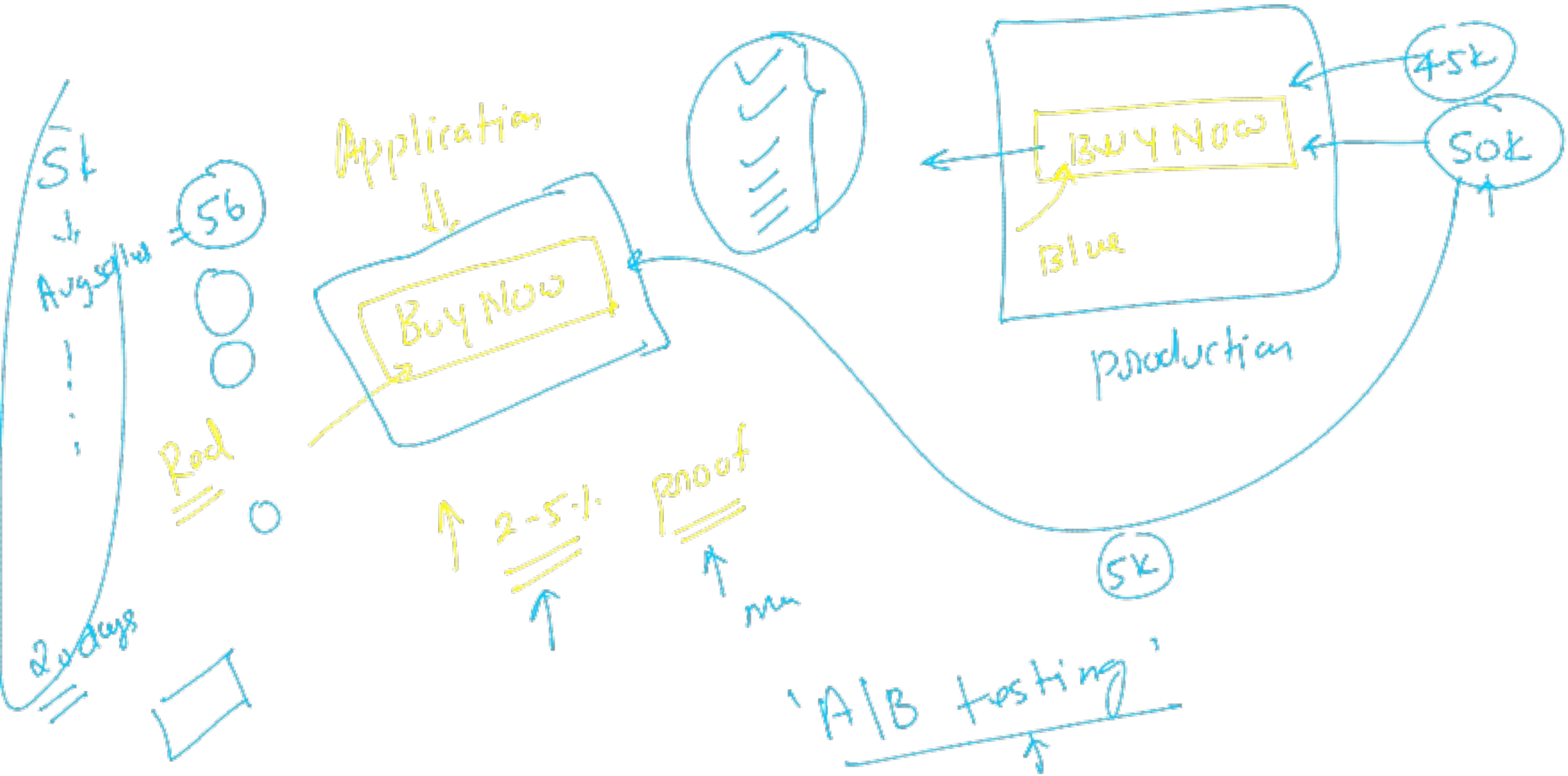
⇒ sum([ ]) / 7

500

500

[- - - -]

Avg hi


5k
Avg
56
Application
Buy Now
Red
2-5-1
proof
Buy Now
Blue
production
45k
50k
5k
'A/B testing'

$S_1 \rightarrow \bar{x}_1$

$S_2 \rightarrow \bar{x}_2$

$S_3 \rightarrow \bar{x}_3$

$\vdots$

$S_n \rightarrow \bar{x}_n$

Avg $\rightarrow$

$\mu = \bar{x}$

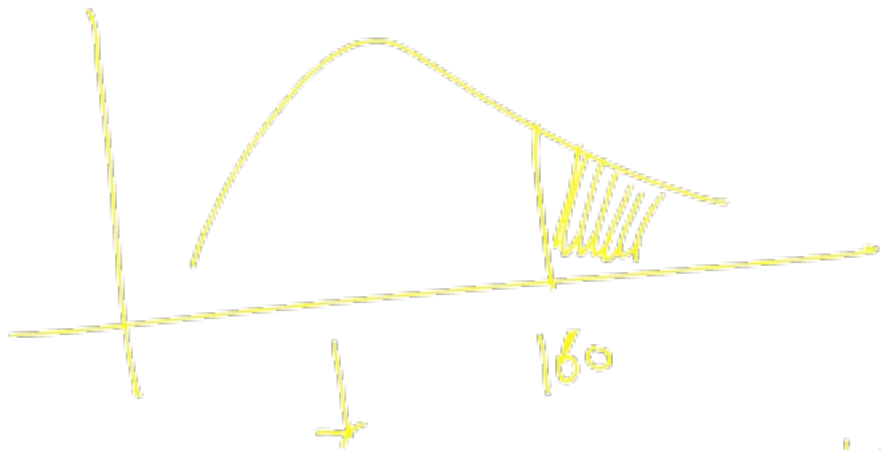
$S_1 \cup S_2$

$\parallel$

population

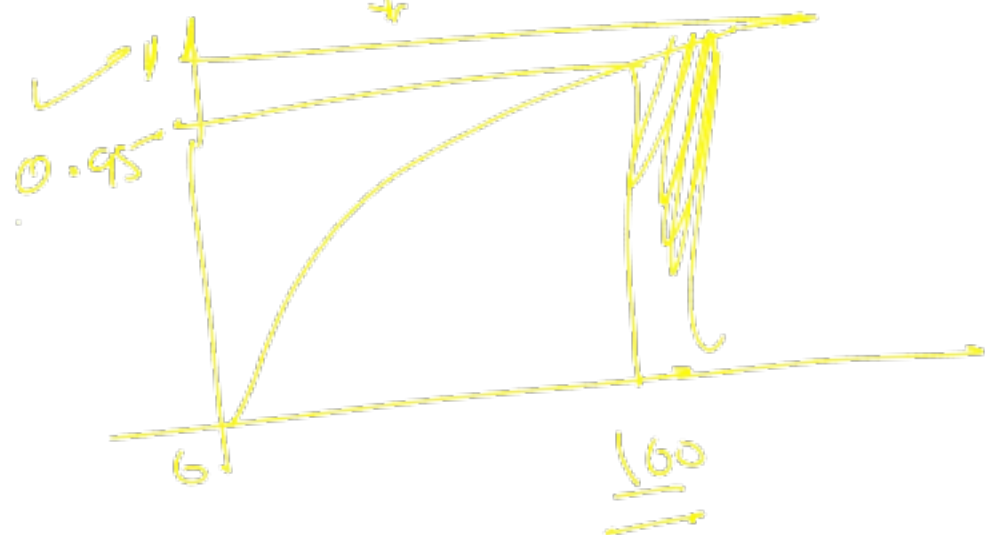
PDF

$N_i \rightarrow$



$\int f(x)\,dx$

CDF



S.I.

$h = \{165, 155, 154, \ldots\} \rightarrow$ Table

$P(h > 160) \rightarrow (0, 1)$

$\int_{-\infty}^{+\infty} f(x)\,dx = 1$

C.R.V

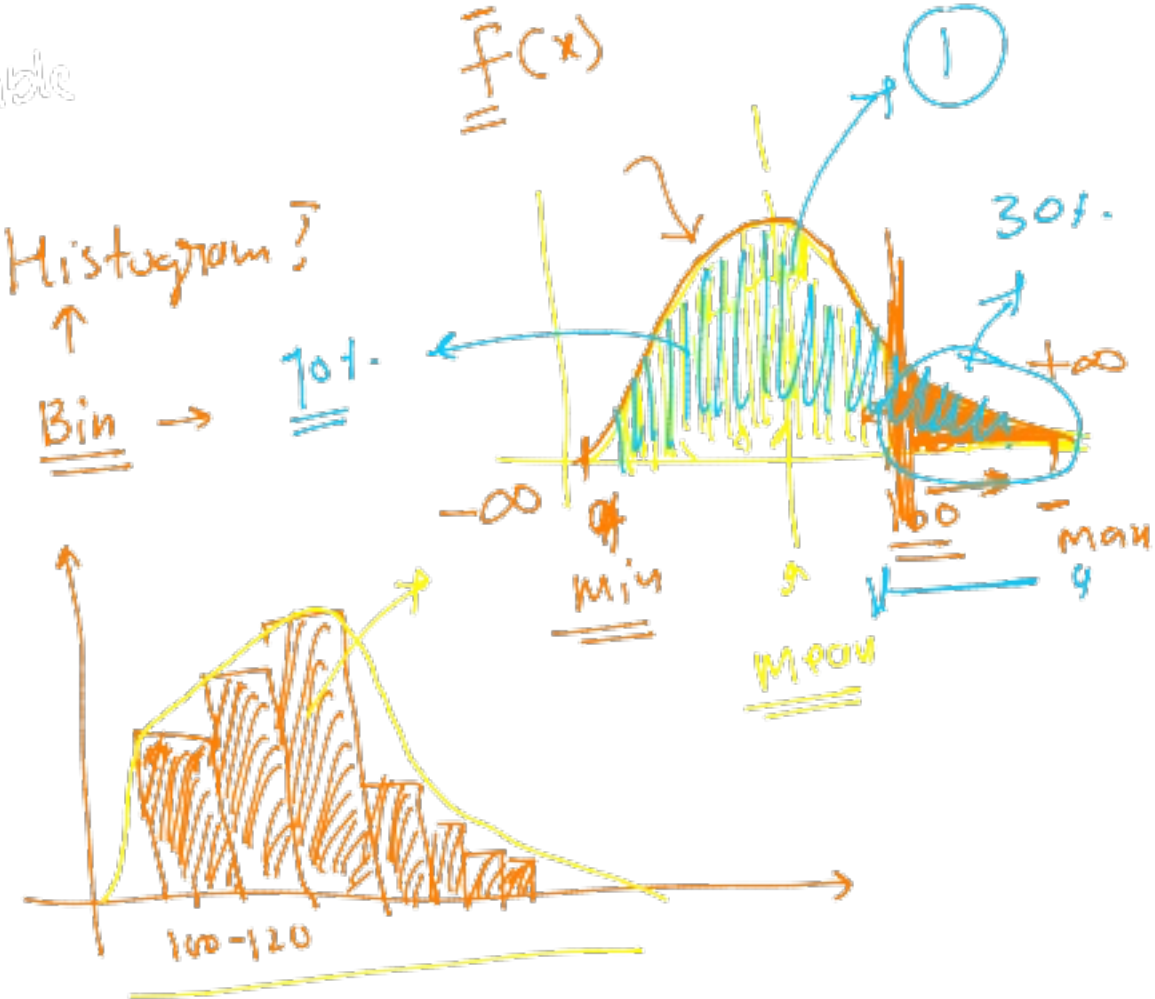
P.d.f

$\int_{-\infty}^{\infty} f(x)\,dx = 1$

Count

Histogram?

Bin → 70%



$$P(x > 160) = \int_{160}^{\infty} f(x)\,dx$$

$$\Downarrow$$

$$\underline{\underline{0.30}}$$

(0-1)

$$\int_{-\infty}^{\infty} f(x) = 1$$

known distribution

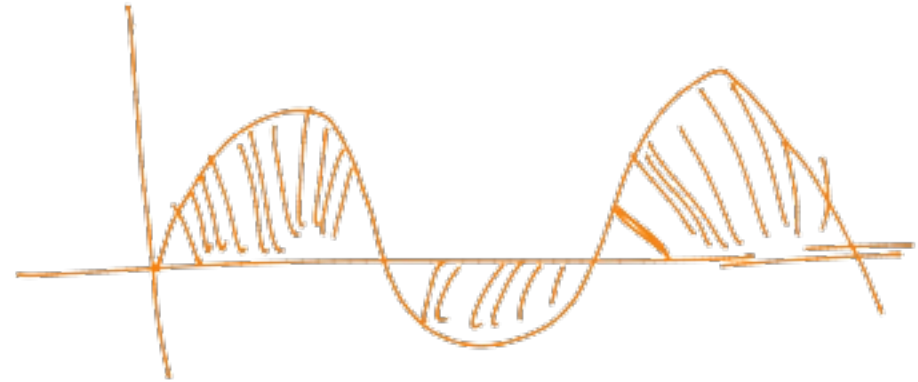
$$P(x > 160) = \int_{160}^{\infty} f(x)\,dx$$

'unknown'?

h_i

$f(x)$

$f(x)$



$f(x) = \sin x$

$$\boxed{\int_{-\infty}^{\infty} \sin x\,dx}$$

Gaussian distribution | Normal distribution

saking $X \sim$ C.R.V.

$f(x)$

How?

$f(x)$

II param $\rightarrow$ $\mu(x)$, $var(x)$

$$P(X=x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left\{ \frac{(x-\mu)^2}{2\sigma^2} \right\}$$

Look?

plot

Sub

$x \to \begin{cases} \mu = 0 \\ \sigma^2 = 1 \\ \sigma = 1 \end{cases}$ say)

$$\frac{1}{\sqrt{2\pi}} \exp\left\{ -\frac{x^2}{2} \right\}$$

Remove

$\exp(-x^2) \to e^{-x^2}$

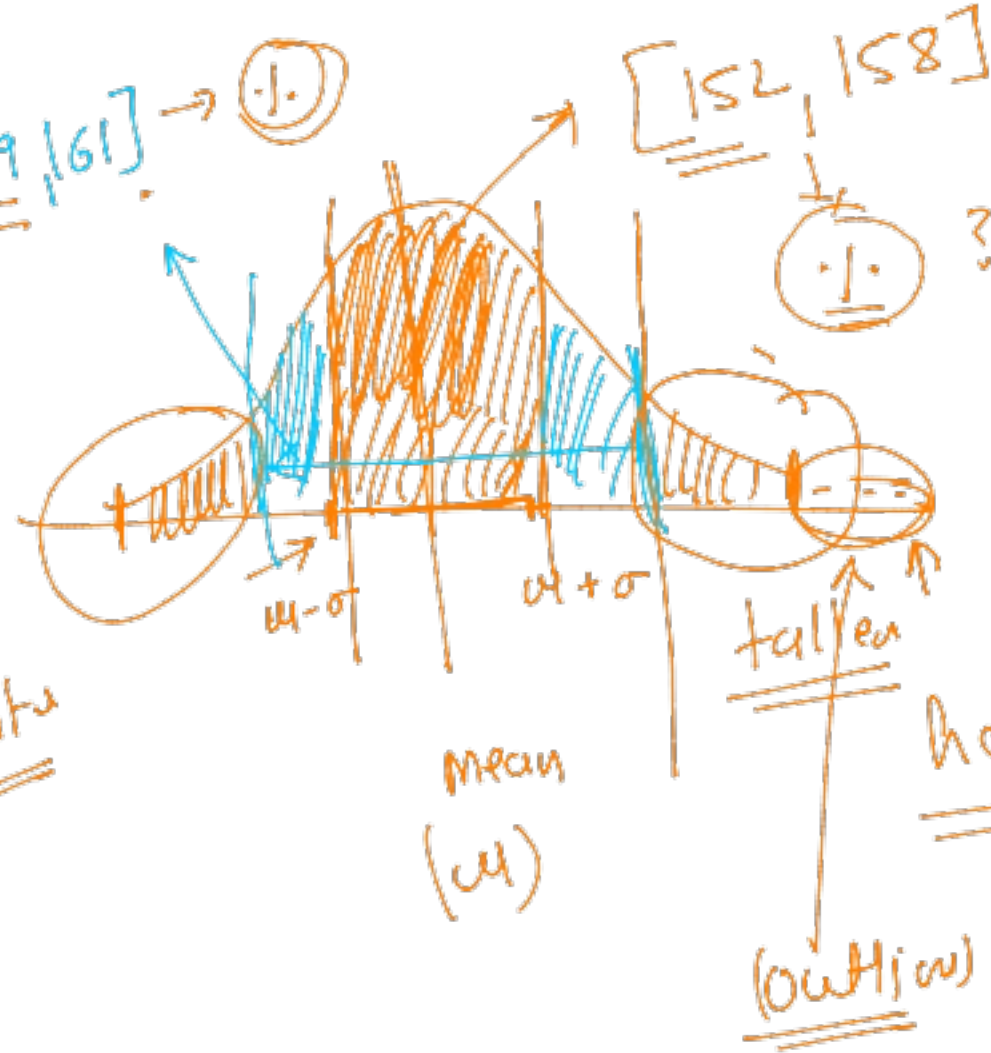
h_i

$\downarrow$

$\mu(x)$

$Var(x)$

plot



Avg $h = 155$

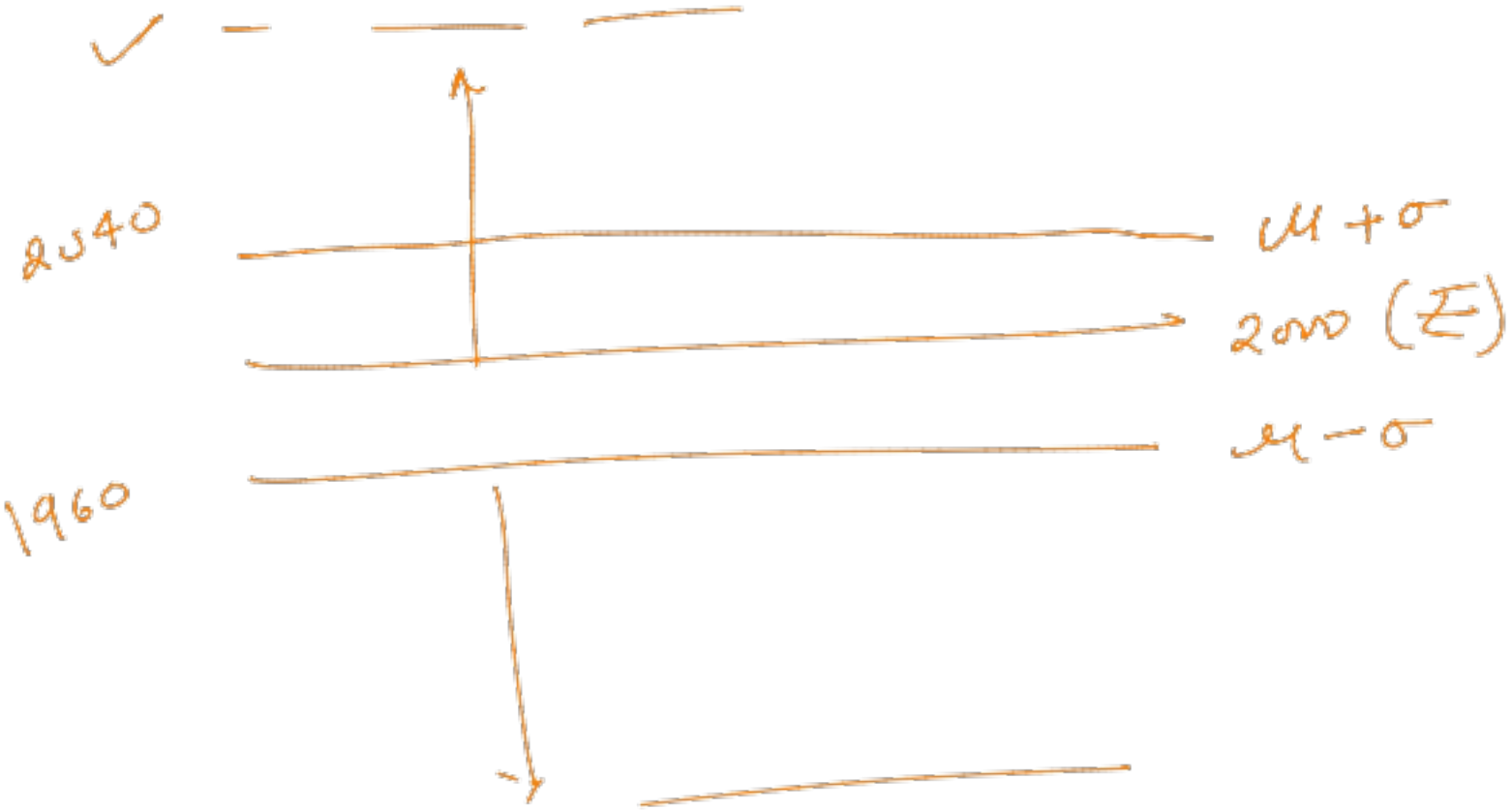
$\sigma = 3$

$[155 - 3(1), 158]$

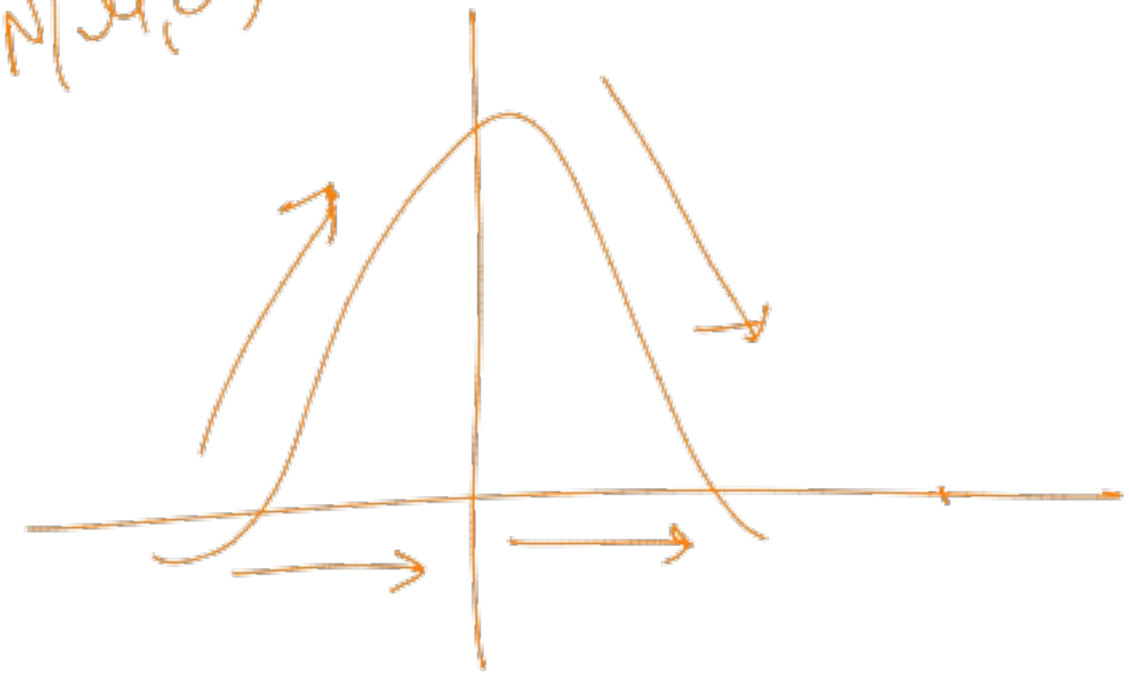
$[152, 158]$

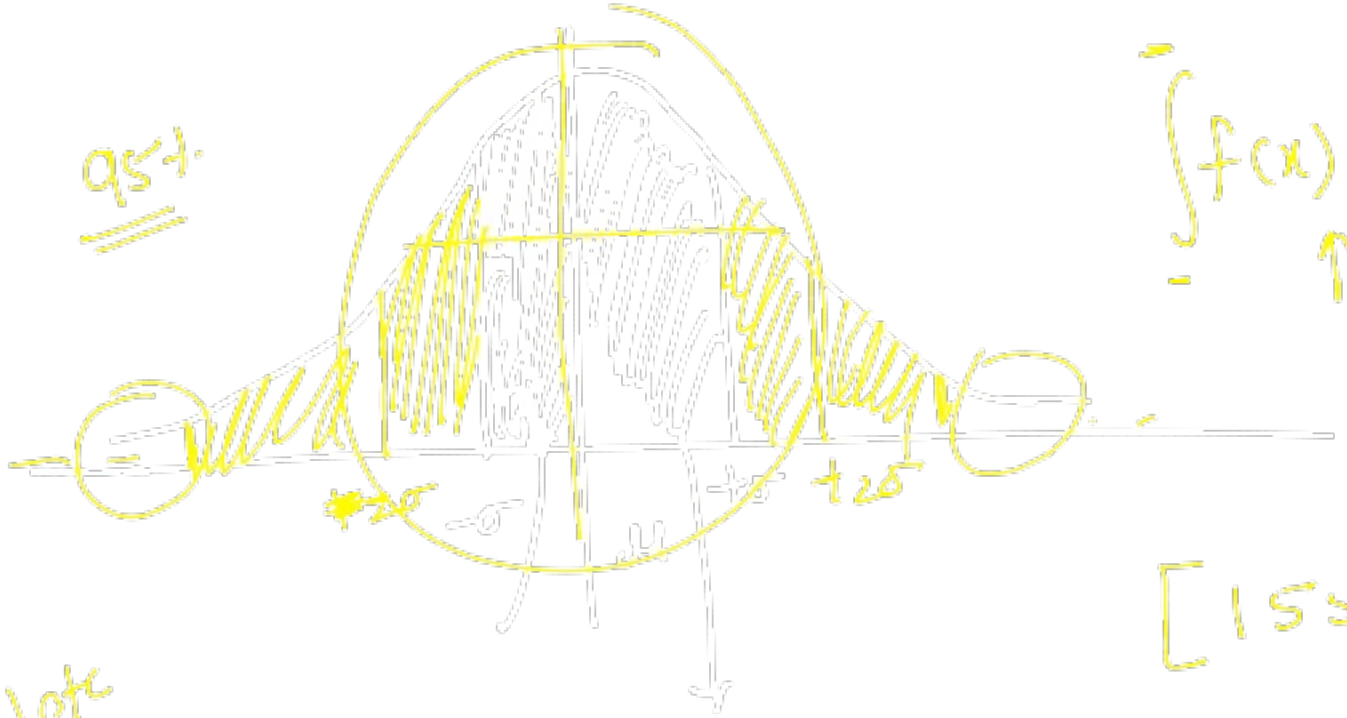
$h_i \to N$

The
↓
2000₹
↓
deviatio
$\sigma = 40₹$



$\theta \to N(\mu, \sigma)$





$\int_{-}^{-} f(x)\,dx$

library

[155, 161]

Train multiple times

Normal

$$\begin{tikzcd} \mathbb{R} \arrow[r, "\pm"] \arrow[d] & \mathbb{N}^{\mathbb{D}} \\ \mathbb{C} \arrow[ur] & \end{tikzcd}$$

CDF

↓

p.f.n ← [commutative distribution / density]